Universitat de les Illes Balears

MASTER'S THESIS

UNTITLED THESIS

Frank William Hammond Espinosa

Master's Degree in Intelligent Systems (MUSI)
Specialization in Artificial Intelligence.

Centre for Postgraduate Studies

Academic year 2024/25

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Keywords:

Underwater imaging, Applied mathematics

Tutor(s): Julia Navarro Oliver and Ana Belén Petro Balaguer.

Untitled thesis

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Thesis of the Master's degree in Intelligent Systems (MUSI)
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***Abstract*—This is a very abstract abstractionary abstract.**

I. Introduction

Blabla cites SOTA etc avorrit.

II. Mathematical model

A. Theoretical grounding and notation

Three representations of images will be used in this work, the equivalences between which will be described below.
The most common representation is that of a $2D$ or $3D$ array, depending on whether the image is grayscale or colored.
Explicar:

1) Molt breument espais L^p, modelització d'imatges com a funcions a L^p.
2) Resum molt resumit d'anàlisi convexa: conjugada convexa; diferenciabilitat de Gateaux i principi de Pascal.
3) Resum molt resumit d'operador proximal i algorisme de Chambolle-Pock.

B. Physical model and formalization of the problem

1) Underwater Image Formation Model: PENDENT: Explicar un mínim d'on surt aquella equació (UIFM) i posar cites.
Let I be the captured image (i.e., the input), J the scene radiance (i.e., the desired image), t the transmission map, g the point-spread funcion and B the background light. According to the UIFM, for each color channel $c \in \{r, g, b\}$ and pixel x,

$$I^c(x) = (J^c t)(x) + B^c(1 - t(x)) + ((J^c t) * g^c)(x) + \xi^c(x), \tag{1}$$

where $\xi(x)$ is random noise on the pixel x, which will be modelled as white noise (i.i.d. centered gaussians).
Ideally, the desired image J is obtained by inverting the previous formula from the captured image I. However, due to the ill-posedness and general analytical untractability of the problem, it is convenient to split it into two subproblems: define, for each color channel $c \in \{r, g, b\}$, the auxiliary maps given by

$$I_1^c(x) = J^c(x)t(x) + B^c(1 - t(x)) \tag{2}$$

and

$$I_2^c(x) = ((J^c t) * g^c)(x) + \xi^c(x), \tag{3}$$

so that $I = I_1 + I_2$. In fact, in order to avoid inverting the convolution present in eq. (3), traditional techniques simply assume $g \equiv 0$ and get rid of I_2 altogether (CITA) by incorporating the noise into I_1. This, of course, simplifies the problem but results in a biased model which does not capture the full array of physical phenomena involved in the image formation process.
Aquí pentura citar s'article de referència.

C. Formalization of the problem

Our approach consists of firstly estimating B, t and a preliminary version of J with traditional methods. This allows the use of eq. (2) to obtain an estimation of I_1 and take $I_2 = I - I_1$. Finally, we utilize an unfolded variational approach to invert eq. (3). Concretely, since t is fixed, we perform a substitution $Jt \to J$ (which is to be inverted later) and pose the following minimization problem, separately for each channel $c \in \{r, g, b\}$:

$$\operatorname*{argmin}_{(J,g)\in D} E(J, g) = \frac{1}{2}\|J * g - I_2^c\|_2^2 + \mathcal{R}_1(J) + \mathcal{R}_2(g), \tag{4}$$

where $D = L^2(\Omega) \times L^2(\Omega)$, and $\mathcal{R}_1$ and $\mathcal{R}_2$ are some regularization functions in $\mathcal{L}(L^2(\Omega), \mathbb{R})$. The reason behind the use of a generic regularization will become clear when discussing the algorithmic approach to the solution via unfolding.
This problem is still mathematically intractable, since the resulting functional is not convex due to the convolution being bilinear w.r.t. the variable functions. A further simplification is made, whereby the functional is alternatively minimized by each of the two variables separately instead of jointly. To compensate for this simplification, the process is repeated a few times:

$$\begin{cases} g_{n+1}^c = \operatorname*{argmin}_{g\in L^2(\Omega)} E(J_n^c, g) \\ J_{n+1}^c = \operatorname*{argmin}_{J\in L^2(\Omega)} E(J, g_{n+1}^c) \end{cases} \tag{5}$$

Due to the symmetry of the target functional E, both subproblems can be studied and solved via the same schema. To see this, consider the following minimization problem: let $s, y \in L^2(\Omega)$ and $R\colon L^2(\Omega) \to \mathbb{R}$. Fixed those symbols, the idea is to solve for

$$\operatorname*{argmin}_{x\in L^2(\Omega)} F(x) = \frac{1}{2}\|x * y - s\|_2^2 + R(x).$$

Both problems in eq. (5) can be trivially reduced to one of this form by appropriately setting s to I_2, and y and R to either

J_n and $\mathcal{R}_2$ or g_{n+1} and $\mathcal{R}_1$. Thus, studying this problem will yield solutions for both subproblems above.
First, it is necessary to show that it is well-posed, meaning that the operations taking place are well-defined and a unique solution exists. Define the auxiliary linear operator $K\colon L^2(\Omega) \to L^2(\Omega)$ given by

$$Kx = x * y..$$

To see that the codomain of this function is, in fact, L^2, we employ the Young inequality together with the well-known fact that $L^2(\Omega) \subseteq L^1(\Omega)$ for a finitely-measured Ω:

$$\|x * y\|_2 \leq \|x\|_2 \|y\|_1 < +\infty.$$

This also implies the boundedness of K.
Also define $N\colon L^2(\Omega) \to \mathbb{R}$ by $N(a) = \|a - s\|_2^2$, which is clearly strictly convex. In this way, we can express

$$F(x) = N(Kx) + R(x),$$

which is a problem of the same shape of (REFERENCIAR PART DE BACKGROUND MATEMATIC) and satisfies the hypotheses of the Chambolle-Pock algorithm. It yields the following iterates:

$$\begin{cases} x_{n+1} = \operatorname{prox}_{\tau R}(x_n - \tau K^* z_n) \\ \tilde{x}_{n+1} = 2x_{n+1} - x_n \\ z_{n+1} = \operatorname{prox}_{\sigma N^*}(z_n + \sigma K \tilde{x}_{n+1}) \end{cases} \tag{6}$$

Closed-form expressions for K^* and $\operatorname{prox}_{\sigma N^*}$ can be derived analytically (cf. (REF APENDIX)). They are given by

$$\operatorname{prox}_{\sigma N^*}(x) = \frac{x - \sigma s}{\sigma + 1} \quad \text{and} \quad K^* z = z * \overline{y}, \tag{7}$$

where $\overline{y}(x) = y(-x)$.

III. Algorithmic approach

A. First half of the problem (I_1).

Falta: explicar RCP; potser DCP
The way of inverting the formula in eq. (2) is the same as in (CITAR ARTICLE REFERÈNCIA), although the implementation is fully self-made (the original implementation is written in MATLAB and, to the best of the author's knowledge, only publicly available in binary form).
First, the background light is chosen by iteratively splitting the image into four regions. Out of those regions, the one with the highest score, given by the average minus the standard deviation of the pixel values in the region, is selected and split again. The process stops once the selected region is smaller than a given threshold (16 colored pixels in our implementation). Then, the background light of the image is chosen as the pixel closest to white (in euclidean norm) in that small patch. This process is summarized in Algorithm 1.
Falta explicar com se calcula es transmission map i es guided filter. He posat s'algoritme pes transmission map a 2.
Once the background light B has been computed, it is possible to estimate the trasmission map t. We proceed in a similar fashion to (CITAR ART. REF): first, a coarse version of the transmission map t_0 is obtained via

$$t_0(x) = 1 - \min_{y \in \Omega_x} \left(\min \left(\frac{1 - I^r(y)}{1 - B^r}, \frac{I^g(y)}{B^g}, \frac{I^b(y)}{B^b}, \lambda S(y) \right) \right),$$

where the *saturation map* S is defined as 0 for black $(0,0,0)$ pixels and $S = 1 - \frac{\min(I^r, I^g, I^b)}{\max(I^r, I^g, I^b)}$ otherwise.
Then, a fine version of the transmission map, t, is obtained by passing t_0 through a guided filter with I^r as the guide. Cal explicar què fa es guided filter? O basta citar?
Finally, we estimate a first version of J as follows, for each color channel $c \in \{r, g, b\}$:

$$J^c(x) = \frac{I^c(x) - B^c}{\max(t(x), 0.1)} + (1 - B^c) \cdot B^c \tag{8}$$

Algorithm 1: Estimate background light.

```
Data: I
Result: J
img ← I;
while Size(img) > min_size do
    quadrants ← Split(img);
    max_score ← 0;
    for q in quadrants do
        μ ← Mean(q);
        σ ← Std(q);
        score ← μ − σ;
        if score > max_score then
            max_score ← score;
            img ← q ;
        end
    end
end
```

B. Second half of the problem (I_2).

In order to solve the minimization subproblems in eq. (5), the iterative schema in eq. (6) with the formulae in eq. (7) is used, with the aforementioned symbolic substitutions. The missing operator $\operatorname{prox}_{\tau R}$ is unfolded, meaning that it is substituted by a neural network instead of explicitly derived. The architecture and training of said neural network will be discussed later.
The final algorithm used for obtaining an approximate solution to eq. (4) is summarized in Algorithm 3.

C. Implementation

Xerrar de U2Fold, la implementació en pytorch dels algoritmes d'abans i de la UNET (he de mirar bé si lo que faig és realment una UNET pq això de sumar en comptes de concatenar és raro), etc.

IV. Experimentation

V. Conclusions

VI. Appendix

Only if deemed necessary.

References

Algorithm 2: Estimate scene radiance and transmission map.

Data: I, B, patch radius r
Result: J_0, t
; /* Compute coarse transmission map */
$S \leftarrow$ Saturation(I);
Initialize t with the width and height of I;
for *pixel* x **do**
  ; /* Ω_x: patch around x, radius r */
  r_min $\leftarrow \min_{y \in \Omega_x}(1 - I^r[\text{y}])/(1 - B^r)$;
  g_min $\leftarrow \min_{y \in \Omega_x}(I^g[\text{y}])/B^g$;
  b_min $\leftarrow \min_{y \in \Omega_x}(I^b[\text{y}])/B^b$;
  s_min $\leftarrow \min_{y \in \Omega_x}(S[\text{y}]) \cdot \lambda$;
  $t[x] \leftarrow 1 - \min$(r_min, g_min, b_min, s_min)
end
; /* Refine transmission map */
$t \leftarrow GuidedFilter(I^r, t)$;
; /* Estimate J_0 */
for *channel* c **do**
  $J_0^c \leftarrow (I^c - B^c)/(max(t, 0.1)) + (1 - B^c) \cdot B^c$
end
return J_0, t

Algorithm 3: Solve second half of the problem.

Data: I_2, t, J_0
Data: σ, τ
Result: J
$J \leftarrow J_0$;
Initialize g;
Initialize dual variables $\tilde{g}$, $\tilde{J}$;
for $n = 1$ **to** G **do**
  ; /* Fix J, estimate g. */
  $\overline{J} \leftarrow DoubleFlip(J)$;
  for $s = 1$ **to** S **do**
    $tmp \leftarrow NeuralNet_{ns}^{g}\left(g - \tau \cdot \tilde{g} * \overline{J}\right)$;
    $g \leftarrow 2 \cdot tmp - g$;
    $\tilde{g} \leftarrow (\tilde{g} + \sigma \cdot (g * J) - \sigma \cdot I_2)/(\sigma + 1)$;
  end
  ; /* Fix g, estimate J. */
  $\overline{g} \leftarrow DoubleFlip(g)$;
  for $s = 1$ **to** S **do**
    $tmp \leftarrow NeuralNet_{ns}^{J}\left(J - \tau \cdot \left(\tilde{J} * \overline{g}\right)\right)$;
    $J \leftarrow 2 \cdot tmp - J$;
    $\tilde{J} \leftarrow (\tilde{J} + \sigma \cdot (J * g) - \sigma \cdot I_2)/(\sigma + 1)$;
  end
end
return J / t